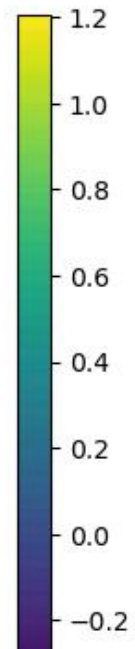
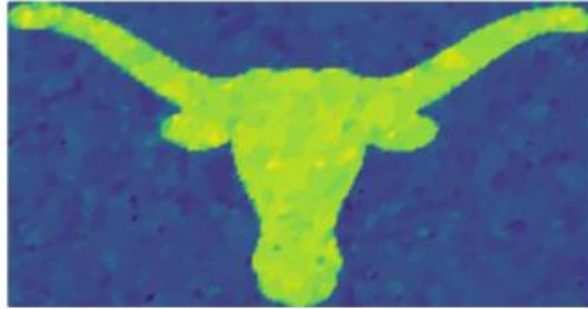


Some select regularization constants and their corresponding outputs are shown below:

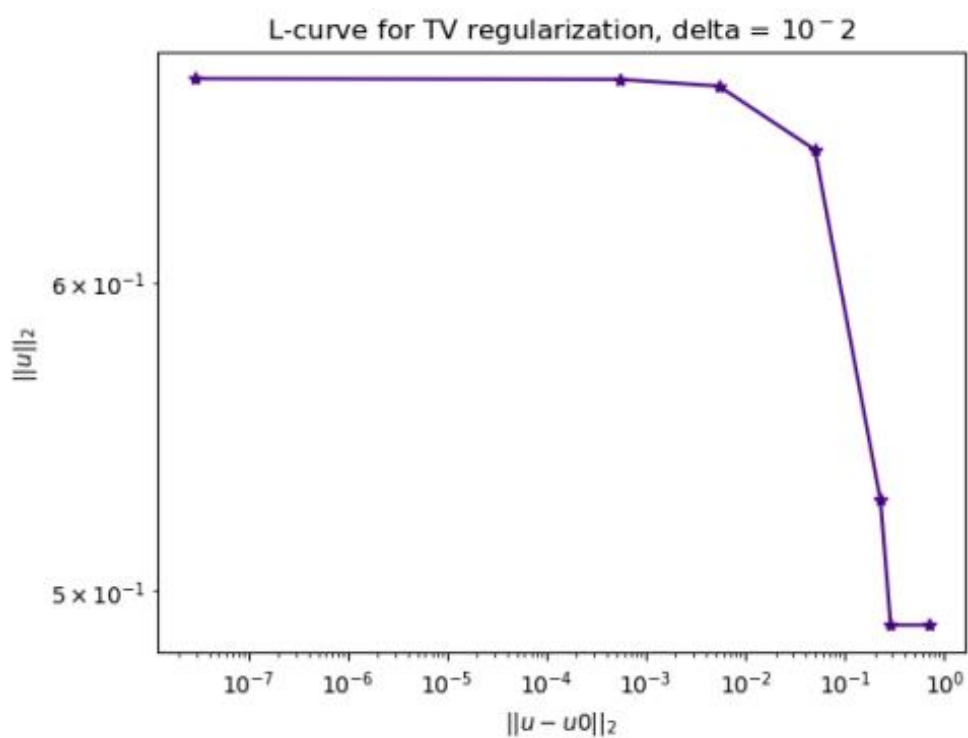
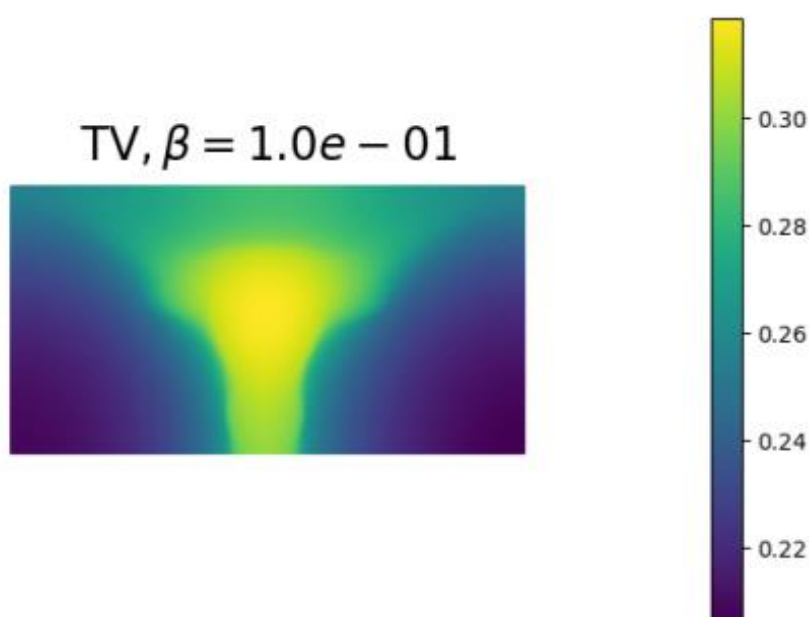
TV, $\text{delta} = 0.01, \beta = 1.0e - 03$



TV, $\text{delta} = 0.01, \beta = 1.0e - 02$



The above graph is the optimal solution (ideal amount of total variation regularization), per visual inspection and the L-curve (shown below).



The optimal beta-delta combination (at least for $\delta = 10^{-2}$) is $\beta = 10^{-2}$.

To check how the number of nonlinear newton iterations changes with delta:

```
In [10]: def nonlinear_it_count(delta):
u = dl.Function(V)
u_hat = dl.TestFunction(V)
u_tilde = dl.TrialFunction(V)

beta = dl.Constant(10**-2)
E = (1/2 * dl.inner(u-u_0, u-u_0) + beta * (ufl.inner(ufl.grad(u), ufl.grad(u))

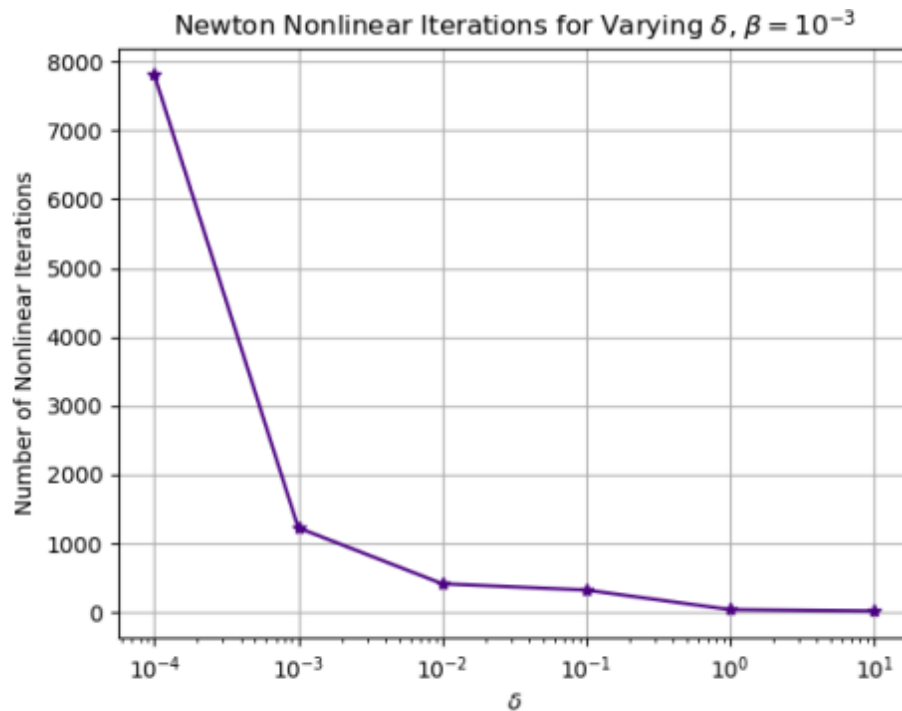
grad = dl.derivative(E, u, u_hat)
H = dl.derivative(grad, u, u_tilde)

solver.solve(E, u, grad, H)

return u, solver.it

nliters = []
deltas = np.logspace(1, -4, 6)
for delta in deltas:
    print(delta)
    _, nliter = nonlinear_it_count(delta = dl.Constant(delta))
    nliters += [nliter]

plt.plot(deltas, nliters, "+-", color = "indigo")
plt.xscale("log")
plt.grid()
plt.title(r"Newton Nonlinear Iterations for Varying $\delta$, $\beta = 10^{-3}$")
plt.xlabel("$\delta$")
plt.ylabel("Number of Nonlinear Iterations")
plt.show()
```



As δ decreases, the number of nonlinear iterations required for convergence increases. This is because the hessian for the newton step becomes increasingly illconditioned; as δ tends 0, the functional R_{TV}^δ becomes non-differentiable when $\nabla u = 0$ (smooth color in sections of the image). As shown in 1b and 1c, the denominator of the first variation of R_{TV}^δ , $(\nabla u \cdot \nabla u + \delta)^{\frac{1}{2}}$, approaches 0; we know this constant will transfer over to the 2nd variatic of R_{TV}^δ , where the hessian entries grow very large in magnitude and become very sensitive to changes in u (thus the hessian is ill conditioned). As the hessian becomes increasingly ill conditioned, the number of nonlinear newton steps for convergence, thus, also increases.

Note in the above graph, beta equals 10^{-2} , not 10^{-3} ; there is a typo here.

46) As shown/proven in 1d), TN regularization overly ²⁵ smoothens/blurs the image, even across sharp boundaries/edges in the image. TV regularization only smoothens in directions orthogonal to the gradient, thus smoothing the high frequency white noise - enhancing image clarity - while still respecting/preserving edge sharpness, resulting in optimal image smoothing.